

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** VALVES FOR CHEMICAL LINES *****			814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.
20										B1	ALL BOLTS ARE METRIC DIA.
20		VC3	CHECK	C01AP76	0.75	8	CHECK 150# RF LTCS VERTICAL BALL TYPE PTFE LINED	0	0	GP1	USE GASKET ONLY WHEN MATING TO A METAL FLANGE SURFACE.
30		VC5	CHECK	C01AP77	0.75	4	CHECK 150# RF LTCS SPRUNG HORIZONTAL BALL TYPE PTFE LINED	0	0		
40										GP2	FOR USE BETWEEN PTFE (RF) AND RUBBER LINED (RF) CONNECTIONS
40			BALL	B01AP39	0.75	2	BALL 150# RF LTCS PTFE LINED FLOATING SOFT FB LO	0	0		
50	U04		BALL	B01AP49	3	4	BALL 150# RF LTCS PTFE LINED FLOATING SOFT RB LO	0	0	GP3	FOR USE BETWEEN PTFE (RF) AND RUBBER LINED (FF) CONNECTIONS.
60	U04		BALL	B01AP50	6	8	BALL 150# RF LTCS PTFE LINED FLOATING SOFT RB GO	0	0		
70							***** VALVES FOR WATER LINES *****			PM6	THIS PIPE IS SUPPLIED CUT TO LENGTH WITH INTEGRAL FLANGES AND LINING. SPECIFY LENGTH OF EACH PIECE.
70											
80											
80		ZA4	GATE	G01AB47	0.75	8	GATE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 HW	0	0	VC3	INSTALL THESE CHECK VALVES IN THE VERTICAL POSITION WITH FLOW UPWARD ONLY.
90											
90		ZA4	BUTTERFLY	UA1B2	10	24	BUTTERFLY 150# RF DOUBLE FLANGED TYPE TRIPLE OFFSET DESIGN FIRE SAFE GEAR OP BS 1400 GR.AB2	0	0	VC5	SPRUNG CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
100											
100		ZA4	GLOBE	L01AB47	0.75	6	GLOBE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 HW	0	0		
110		ZA4	GLOBE	L01AB58	8	8	GLOBE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 GO	0	0		
120										VU9	USE 1/2" NB DUPLEX SS FLANGE, NIPPLES AND BALL VALVE FOR ORIFICE TAP BLOCK VALVES ONLY.
120			CHECK	C01AB78	2	6	CHECK DUAL WAFER LUG 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 THROU BOLTED	0	0		
130	U04		CHECK	C01AB59	8	24	CHECK DUAL DBL FLG 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2	0	0	ZA4	COMPONENT SHALL ONLY BE USED ON WATER SERVICES (WRA, WP, WE)
140							***** PIPE *****				
140											
150											
150		PM6	PIPE	5717206	0.75	2	PIPE SCH XS SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS	0	0		
160		PM6	PIPE	5717205	3	10	PIPE SCH 40 SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS	0	0		
170			PIPE	5717204	12	24	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS	0	0		
180		814	PIPE	5366119	0.5	0.5	PIPE SCH 160 SMLS DUPLEX STL S31803	0	0		
190							***** FITTINGS *****				
190											
200											
200			ELBOW	5717620	0.75	10	ELL 90 DEG 150# FLANGED LTCS PTFE LINED	0	0		
210			ELBOW	5717630	12	24	ELL 90 DEG 150# FLANGED SCH 20 LTCS PTFE LINED	0	0		
220			ELBOW	5717621	0.75	10	ELL 45 DEG 150# FLANGED LTCS PTFE LINED	0	0		
230			ELBOW	5717631	12	24	ELL 45 DEG 150# FLANGED SCH 20 LTCS PTFE LINED	0	0		
240											
240			TEE	5717622	0.75	10	TEE EQUAL 150# FLANGED LTCS PTFE LINED	0	0		
250			TEE	5717632	12	24	TEE EQUAL 150# FLANGED SCH 20 LTCS PTFE LINED	0	0		
260			FLANGE	5717623	0.75	8	TEE REDUCING 150# FLANGED LTCS PTFE LINED	1	10		
270	U04		TEE RED	5717636	4	10	TEE REDUCING 150# FLANGED SCH 20 X SCH 40 LTCS PTFE LINED	12	24		
280			FLANGE	5717633	12	20	TEE REDUCING 150# FLANGED SCH 20 LTCS PTFE LINED	14	24		
290											
290			TEE	5717627	0.75	2	INSTRUMENT TEE 150# FLANGED LTCS PTFE LINED	1	10		
300			TEE	5717637	0.75	2	INSTRUMENT TEE 150# FLANGED SCH 20 X SCH XS LTCS PTFE LINED	12	24		
310											
310			RED CONC	5717624	0.75	8	REDUCER CONCENTRIC 150# FLANGED LTCS PTFE LINED	1	10		
320			RED CONC	5717638	6	10	REDUCER CONCENTRIC 150# SCH 20 X SCH 40 FLANGED LTCS PTFE LINED	12	12		
330			RED CONC	5717634	12	20	REDUCER CONCENTRIC 150# SCH 20 FLANGED LTCS PTFE LINED	14	24		
340			RED ECC	5717625	0.75	8	REDUCER ECCENTRIC 150# FLANGED LTCS PTFE LINED	1	10		
350			RED ECC	5717639	8	10	REDUCER ECCENTRIC 150# SCH 20 X SCH 40 FLANGED LTCS PTFE LINED	12	12		
360			RED ECC	5717635	12	20	REDUCER ECCENTRIC 150# SCH 20 FLANGED LTCS PTFE LINED	14	24		
370							***** FLANGES *****				
370											
380											
380	U04		FLANGE	5717628	0.5	20	REDUCING FLG 150# PTFE LINED LTCS A350 LF2 CL.1	0.75	24		
390											
390	U04		BLIND	5715115	0.5	24	FLG BLIND 150# PTFE LINED LTCS A350 LF2 CL.1	0	0		
400	U04		BLIND	5715130	0.5	0.5	FLG BLIND 300# PTFE LINED LTCS A350 LF2 CL.1	0	0		
410							***** SPACERS *****				
410											
420											
420		GP2	PADDLE	5717618	0.75	12	STANDARD RING SPACER 150# PTFE	0	0		
430		GP3	PADDLE	5717619	0.75	12	STANDARD FULL FACE SPACER 150# PTFE	0	0		
440							***** GASKETS *****				
440											
450											
450		GP1	GASKET	5664201	0.5	24	GASKET 150# RF FLAT RING PTFE	0	0		
460		GP1	GASKET	5664204	0.5	0.5	GASKET 300# RF FLAT RING PTFE	0	0		
470							***** BOLTING *****				
470		B1									
470											
480	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0		
490											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		LTCS PIPING + PTFE LINED. ALL SIZES-FLANGED. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 0 MM	DESIGN : 16 BARG CODE : ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: PTFE LINED LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 150 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	150CP2

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490							***** SPECTACLE BLINDS *****			
500			SPEC BLIND	5749585	0.75	12	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 PTFE LINED AND FACED	0	0	
510			PADDLE	5750024	16	16	PADDLE BLANK 150# RF LTCS A516 GR.70 PTFE LINED AND FACED	0	0	
520			PADDLE	5750243	16	16	PADDLE SPACER 150# RF LTCS A516 GR.70 PTFE LINED AND FACED	0	0	
530							*****MISCELLANEOUS*****			
540	U04		FLG GUARD	5558881	0.5	10	SAFETY SHIELD PTFE SUITABLE FOR 150# FLANGE	0	0	
550							*****PIPE NIPPLES*****			
560	814			0			NIPPLES PBE TO BE MADE FROM PIPE			
570	VU9		NIPPLE	5542170	0.5	0.5	NIPPLE SCH 80S A790 UNS S31803 POE/TOE 3" LONG	0	0	
580	VU9		NIPPLE	5542172	0.5	0.5	NIPPLE SCH 80S A790 UNS S31803 TOE/POE 7 IN LONG	0	0	
590										

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		LTCS PIPING + PTFE LINED. ALL SIZES-FLANGED. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 0 MM	DESIGN : 16 BARG CODE : ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: PTFE LINED LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE:- 40 / 150 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	150CP2